

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML-A10
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Adaptive Interview Intelligence Platform Using Multi-Agent AI and Retrieval-Augmented Generation.

Abstract

The recruitment process has become increasingly competitive, requiring candidates to possess strong technical knowledge, communication skills, and problem-solving abilities. However, many students and job seekers lack access to personalized interview preparation and meaningful feedback. Existing methods, such as static question banks and conventional mock interview platforms, provide generic practice sessions without adapting to individual skills, learning progress, or target job roles. Consequently, candidates often struggle to identify their weaknesses and improve effectively before attending real interviews.

This project is designed for students, graduates, job seekers, educational institutions, placement training centers, and recruiters seeking an intelligent interview preparation solution. Existing platforms, including Interview Warmup by Google, Pramp, Interviewing.io, and LeetCode, offer mock interviews and practice questions but provide limited personalization, explainable feedback, and adaptive learning. These limitations create the need for a more intelligent and user-centric interview preparation platform.

The proposed **Adaptive Interview Intelligence Platform Using Multi-Agent AI and Retrieval-Augmented Generation** integrates multiple AI agents to perform specialized tasks such as resume analysis, adaptive interview question generation, candidate evaluation, knowledge retrieval, personalized recommendations, and performance reporting. Retrieval-Augmented Generation (RAG) retrieves relevant information from trusted knowledge sources before generating responses, improving accuracy and reducing AI hallucinations. Adaptive learning continuously adjusts interview questions and recommendations based on each candidate's performance, providing a personalized learning experience.

The scope of this project includes AI-based resume analysis, technical and HR mock interviews, adaptive question generation, intelligent answer evaluation, explainable feedback, personalized learning recommendations, and performance analytics through a web-based platform. The project does not replace human interviewers or make final recruitment decisions.

The expected outcome is an intelligent, scalable, and research-oriented platform that enhances interview readiness through personalized AI guidance. The success of the project will be measured by improved candidate performance, accurate AI-based evaluation, high user satisfaction, and the ability of at least **90% of users** to complete personalized interview preparation independently. This project aims to bridge the gap between traditional interview preparation methods and next-generation AI-powered learning systems, providing significant academic, industrial, and commercial value.

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References (Category specific)

1. EduMSRA: A Multi-Source Educational Research Agent Integrating Retrieval-Augmented Generation and Model Context Protocol for Adaptive Intelligent Tutoring Systems - https://www.mdpi.com/2076-3417/16/9/4400?utm_source=chatgpt.com
2. Bringing Generative AI to Adaptive Learning in Education - https://doi.org/10.48550/arXiv.2402.14601?utm_source=chatgpt.com
3. A Survey on Retrieval-Augmented Text Generation for Large Language Models - https://arxiv.org/abs/2404.10981?utm_source=chatgpt.com
4. Retrieval-Augmented Generation for AI-Generated Content: A Survey - https://arxiv.org/abs/2402.19473?utm_source=chatgpt.com
5. A Comprehensive Survey of Retrieval-Augmented Generation (RAG): Evolution, Current Landscape and Future Directions - https://doi.org/10.48550/arXiv.2410.12837?utm_source=chatgpt.com

Project Team Members

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Project Guide

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Department of CSE - Artificial Intelligence & Machine Learning

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.